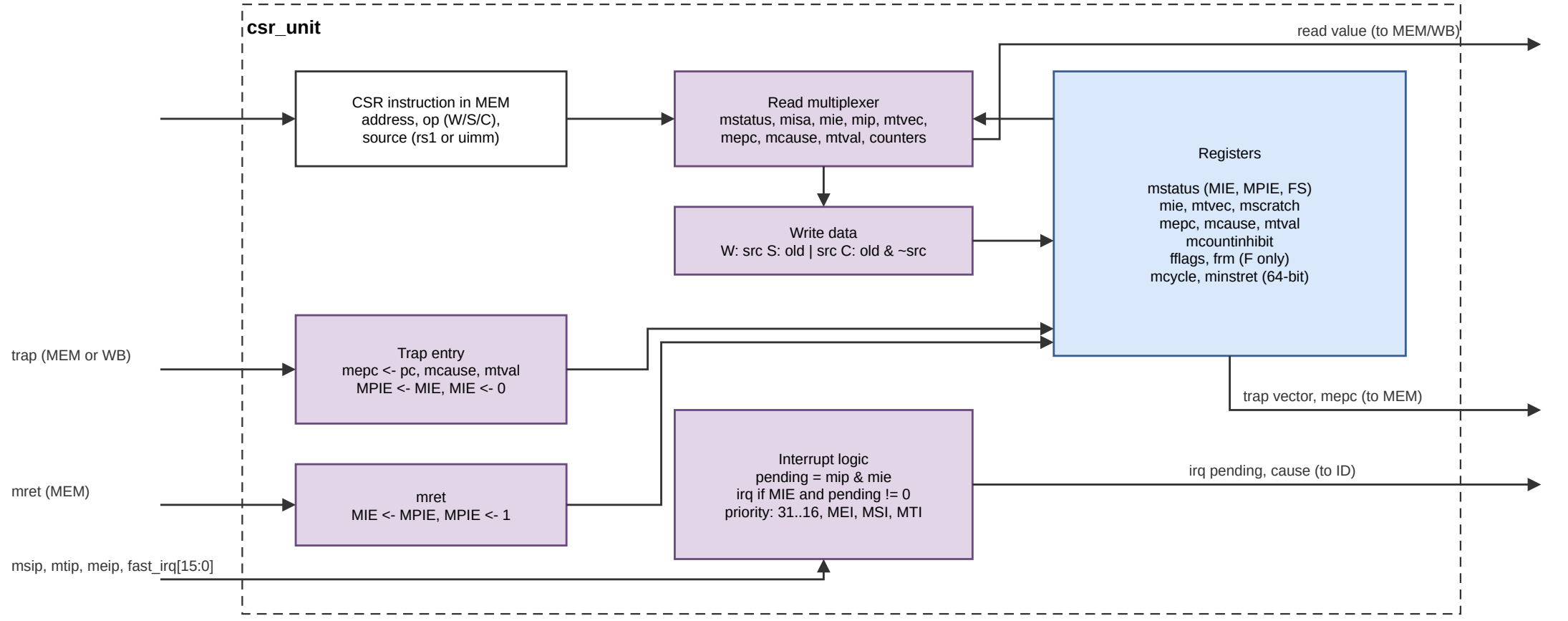
csr_unit
CSR instruction in MEM
address, op (W/S/C),
source (rs1 or uimm)
Read multiplexer
mstatus, misa, mie, mip, mtvec,
mepc, mcause, mtval, counters
read value (to MEM/WB)
Registers
mstatus (MIE, MPIE, FS)
mie, mtvec, mscratch
mepc, mcause, mtval
mcountinhibit
fflags, frm (F only)
mcycle, minstret (64-bit)
Write data
W: src S: old | src C: old & ~src
trap (MEM or WB)
Trap entry
mepc <- pc, mcause, mtval
MPIE <- MIE, MIE <- 0
trap vector, mepc (to MEM)
mret (MEM)
mret
MIE <- MPIE, MPIE <- 1
Interrupt logic
pending = mip & mie
irq if MIE and pending != 0
priority: 31..16, MEI, MSI, MTI
irq pending, cause (to ID)
msip, mtip, meip, fast_irq[15:0]